The sharp L^1 Poincaré inequality on the cube

This explanation is based on Tomar's recent preprint [2], which uses the semigroup identity of Ivanisvili–van Handel–Volberg [1]. We state only the three ingredients and the final deduction. For each lemma, one can ask AI to explore alternative or simpler proofs.

For the uniform measure on $\{-1,1\}^n$, put $D_j f(x) = (f(x) - f(x^{(j)}))/2$ and $|\nabla f| = (\sum_j (D_j f)^2)^{1/2}$, where $x^{(j)}$ flips the jth coordinate of x. The sharp dimension-independent inequality is

$$\mathbb{E}|f - \mathbb{E}f| \le \sqrt{\frac{\pi}{2}}\, \mathbb{E}|\nabla f|.$$

Lemma 1 (semigroup reduction). For every $f : \{-1,1\}^n \to \mathbb{R}$,

$$\mathbb{E}|f - \mathbb{E}f| \le \mathbb{E}J(\nabla f), \qquad J(a) = \frac{1}{2}\int_0^1 \frac{\mathbb{E}\left|\sum_{j=1}^n a_j (B_j(q) - q)\right|}{q(1-q)}\, dq,$$

where, for each q, the variables $B_j(q)$ are independent Bernoulli variables of parameter q.

Comment. This follows from the semigroup identity in [1, Lemma 2.1 and Remark 2.2] and is the main input from that paper, which also resolved Enflo's conjecture that Rademacher type and Enflo type coincide for Banach spaces.

Lemma 2 (removing signs; Tomar). For every $a \in \mathbb{R}^n$,

$$J(a) \le J(|a|), \qquad |a| = (|a_1|, \dots, |a_n|).$$

Comment. This comparison was first proved in Tomar's preprint [2, Theorem 4.1]. Its proof is technical and uses Fourier analysis.

Lemma 3 (positive coefficients; Tomar). If $b_j \ge 0$ for every j, then

$$J(b) = \int_0^\infty \left[1 - \prod_{j=1}^n (1 + b_j t) e^{-b_j t}\right] \frac{dt}{t^2}.$$

Comment. This identity also appears in Tomar's preprint [2, Theorem 3.6]. It admits different proofs, including a short application of the residue theorem.

Deduction of the inequality. For $x \ge 0$, $\log(1+x) \ge x - x^2/2$. Thus Lemma 3 gives, for $b \ge 0$,

$$J(b) \le \int_0^\infty \frac{1 - e^{-t^2 \|b\|_2^2/2}}{t^2}\, dt = \sqrt{\frac{\pi}{2}}\, \|b\|_2,$$

where the equality follows by integration by parts and the Gaussian integral. By Lemma 2, the same bound holds for every $a \in \mathbb{R}^n$. Lemma 1 therefore gives

$$\mathbb{E}|f - \mathbb{E}f| \le \mathbb{E}J(\nabla f) \le \sqrt{\frac{\pi}{2}}\, \mathbb{E}|\nabla f|.$$

References

[1] P. Ivanisvili, R. van Handel and A. Volberg, *Rademacher type and Enflo type coincide*, Ann. of Math. (2) **192** (2020), 665–678. doi:10.4007/annals.2020.192.2.8.

[2] T. S. Tomar, *The L^1 Poincaré constant of the Hamming cube*, preprint. Online PDF.

Paata Ivanisvili
Department of Mathematics, University of California, Irvine
pivanisv@uci.edu

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